

Image Generation using stable diffusion & Comfy UI

A Project Report

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by

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ABSTRACT

In recent years, **AI-powered image generation** has gained significant attention, revolutionizing industries like digital art, media, and content creation. This project, "**Image Generation Using Stable Diffusion and ComfyUI**," explores the fundamentals of **Stable Diffusion**, an advanced generative model that transforms noise into detailed images using a guided iterative process. Conducted as part of this study, we delve into **U-Net architecture, denoising autoencoders, and latent diffusion models** to understand how AI generates high-quality visuals.

The primary objective of this project is to **investigate the working principles of Stable Diffusion, analyze the role of U-Net in noise prediction, and evaluate how conditioning techniques improve image quality**. To achieve this, we follow a structured methodology consisting of:

1. **Understanding Diffusion Models** – Examining the forward and reverse diffusion process to understand how images are generated from random noise.
2. **Exploring U-Net Architecture** – Analyzing how the U-Net model predicts noise and refines image details during the reverse diffusion process.
3. **Implementation using ComfyUI** – Utilizing **ComfyUI**, a modular workflow-based interface, to experiment with Stable Diffusion and generate images under different conditions.
4. **Text-to-Image Generation** – Evaluating how textual prompts influence the generation process and how conditioning techniques like CLIP guidance enhance image fidelity.
5. **Performance Evaluation** – Assessing image quality using **CLIP similarity scores** and other benchmarking techniques.

Key findings indicate that **Stable Diffusion efficiently generates diverse and high-resolution images** while preserving semantic coherence. By leveraging **U-Net's noise prediction capability**, the model refines details at each step, resulting in highly realistic visuals. ComfyUI, with its **node-based workflow**, allows greater control over the image generation process, making it a powerful tool for AI artists and developers.

In conclusion, this project demonstrates how **Stable Diffusion, U-Net, and ComfyUI collectively contribute to AI-based image generation**. The study also opens pathways for future advancements in **video diffusion, interactive AI art, and real-time generative models**.

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CHAPTER 1

Introduction

1.1 Problem Statement:

AI-powered **image generation** has gained immense popularity with the rise of **Stable Diffusion**, an advanced model capable of generating high-quality visuals from random noise. However, despite its effectiveness, **understanding how Stable Diffusion works, including its core components like U-Net and latent diffusion, remains complex**. Many users rely on pre-trained models without fully grasping the underlying principles, limiting their ability to fine-tune or innovate upon existing architectures.

Additionally, traditional image-generation models often face **challenges such as high computational cost, lack of user control, and inconsistencies in generated outputs**. The emergence of **ComfyUI**, a powerful workflow-based interface for Stable Diffusion, allows users to manipulate and optimize the generation process with greater flexibility.

This project aims to address these gaps by:

- **Exploring the working of Stable Diffusion** and how it generates images step-by-step.
- **Understanding the role of U-Net architecture** in denoising and refining images.
- **Implementing ComfyUI** to visualize and customize image generation workflows.
- **Evaluating different conditioning techniques** to improve the quality and control of generated images.

Significance of the Problem:

By analyzing these aspects, this study provides insights into **how AI-based diffusion models generate images** and how they can be improved for better creativity and efficiency.

1.2 Motivation:

Why Was This Project Chosen?

1. **Growing Interest in AI-Generated Art** – AI-generated images are transforming creative fields like digital art, game design, and content creation. Understanding **Stable Diffusion** helps unlock new possibilities in **AI-driven creativity**.
2. **Exploring Core AI Concepts** – This project provides hands-on experience in **deep learning, generative AI, and U-Net models**, strengthening foundational knowledge in AI research.
3. **Understanding ComfyUI for Better Control** – Unlike traditional models with limited user interaction, **ComfyUI allows a modular, workflow-based approach** that enhances flexibility in AI image generation.
4. **Bridging the Gap Between Theory and Implementation** – Many studies focus on using **pre-trained diffusion models**, but this project aims to **explore their internal mechanics** for a deeper understanding.

1.3 Objective:

The primary objective of this project is to **understand and implement image generation using Stable Diffusion and ComfyUI.**

Specific Objectives:

1. **Study the Stable Diffusion Process** – Analyze **forward and reverse diffusion** and how images emerge from noise.
2. **Understand U-Net's Role** – Investigate **how U-Net functions in noise prediction and refinement** during the diffusion process.
3. **Implement Image Generation Using ComfyUI** – Experiment with **node-based workflows** for generating and controlling images.
4. **Optimize Prompt-Based Generation** – Evaluate **how different text inputs influence image outputs** using **CLIP guidance**.
5. **Assess Model Performance** – Use techniques like **CLIP similarity scoring** to evaluate the quality of generated images.

By achieving these objectives, this project provides a **comprehensive understanding of AI-driven image generation.**

1.4 Scope of the Project:

Key Areas Covered:

- ✓ **Stable Diffusion Model:** Understanding its step-by-step image generation process.
- ✓ **U-Net Architecture:** Exploring its function in refining AI-generated images.
- ✓ **ComfyUI Implementation:** Using workflow-based AI tools to manipulate image outputs
- ✓ **Text-to-Image Conditioning:** Analyzing how text prompts influence generated images
- ✓ **Evaluation Metrics:** Measuring the accuracy and quality of generated outputs.

Limitations of the Project:

- ✓ **Computational Constraints** – Running diffusion models requires high-end GPUs for efficient performance.
- ✓ **Training Complexity** – Fine-tuning a Stable Diffusion model from scratch is beyond the scope of this project.
- ✓ **Subjectivity in Image Quality** – Evaluating AI-generated images can be subjective and depends on human interpretation.

This project aims to balance **practical implementation** with **theoretical understanding**, providing a solid foundation for future research in AI-based image generation.

CHAPTER 2

Literature Survey

2.1 Review relevant literature or previous work.

Generative AI models have evolved significantly over the past decade, with **diffusion models** emerging as one of the most powerful techniques for **text-to-image synthesis, video generation, and creative AI applications**. This section reviews key research contributions in this field, focusing on **Stable Diffusion, U-Net architecture, and advancements in AI-based image generation**.

Evolution of Generative Models in AI

Generative models have transitioned from early approaches like **GANs (Generative Adversarial Networks)** to **autoregressive models (PixelCNN, PixelRNN)** and more recently to **diffusion models**. Each advancement has improved **image fidelity, diversity, and control over the generation process**.

- **GANs (Goodfellow et al., 2014)** – Early models like DCGAN (Radford et al., 2015) introduced **adversarial training**, where two networks (generator and discriminator) competed to generate realistic images. However, GANs struggled with **mode collapse** and required extensive training.
- **VAEs (Kingma & Welling, 2013)** – Variational Autoencoders helped in **image reconstruction** but lacked fine-grained detail control.
- **Diffusion Models (Sohl-Dickstein et al., 2015)** – Introduced the idea of **iterative denoising**, which was later improved with **latent diffusion models** for high-resolution synthesis.

2.2 Existing Models and Techniques for Image Generation

▪ Diffusion Models

Diffusion models work by **gradually corrupting an image with Gaussian noise (forward diffusion process) and then learning to reverse this process (reverse diffusion process) to reconstruct a realistic image**.

The foundational work on diffusion models was introduced by **Sohl-Dickstein et al. (2015)**, who proposed a **Markov chain framework** for stepwise denoising. This idea was refined in **DDPM (Denoising Diffusion Probabilistic Models)** by **Ho et al. (2020)**, which improved training efficiency and image quality.

Stable Diffusion: A Latent Diffusion Model (Rombach et al., 2022)

Unlike earlier models that worked on pixel space, **Stable Diffusion operates in latent space** using a pre-trained **variational autoencoder (VAE)** to encode and decode images. This allows **faster and more memory-efficient image generation**.

- **U-Net Architecture in Image Generation**

U-Net was originally designed for **medical image segmentation (Ronneberger et al., 2015)** but has been widely adopted in **diffusion models for noise prediction**.

How U-Net Works in Stable Diffusion:

- It **takes noisy images as input** and predicts the **noise residual**, helping in gradual refinement during denoising.
- Uses **skip connections** to retain fine details lost during convolutional downsampling.
- Works in conjunction with **attention layers** to enhance object-specific features.

2.3 Gaps in Existing Models and How This Project Addresses Them

- **Lack of Real-Time Generation** – GANs are faster, but Stable Diffusion takes multiple steps for refinement. Future research can focus on **accelerating denoising steps**.
- **Computational Challenges** – Stable Diffusion requires high-end GPUs. This project explores **optimization techniques like reducing diffusion steps** and **using ComfyUI's workflow-based approach** to manage memory efficiently.
- **Limited Artistic Control in Text-to-Image Models** – While Stable Diffusion improves **text fidelity**, manual prompt engineering is still needed. ComfyUI provides **workflow-based fine-tuning** for better user control.
- **Challenges in Multi-Modal Image Synthesis** – Current models struggle with **video generation and interactive content creation**. Future research can explore **Stable Video Diffusion** for AI-powered motion synthesis.

CHAPTER 3

Proposed Methodology

3.1 System Design

The system design for this project follows a structured pipeline for Stable Diffusion-based image generation using ComfyUI. The methodology ensures a comprehensive understanding of the diffusion process, U-Net denoising, text-conditioning, and image synthesis.

3.1.1 ComfyUI workflow:

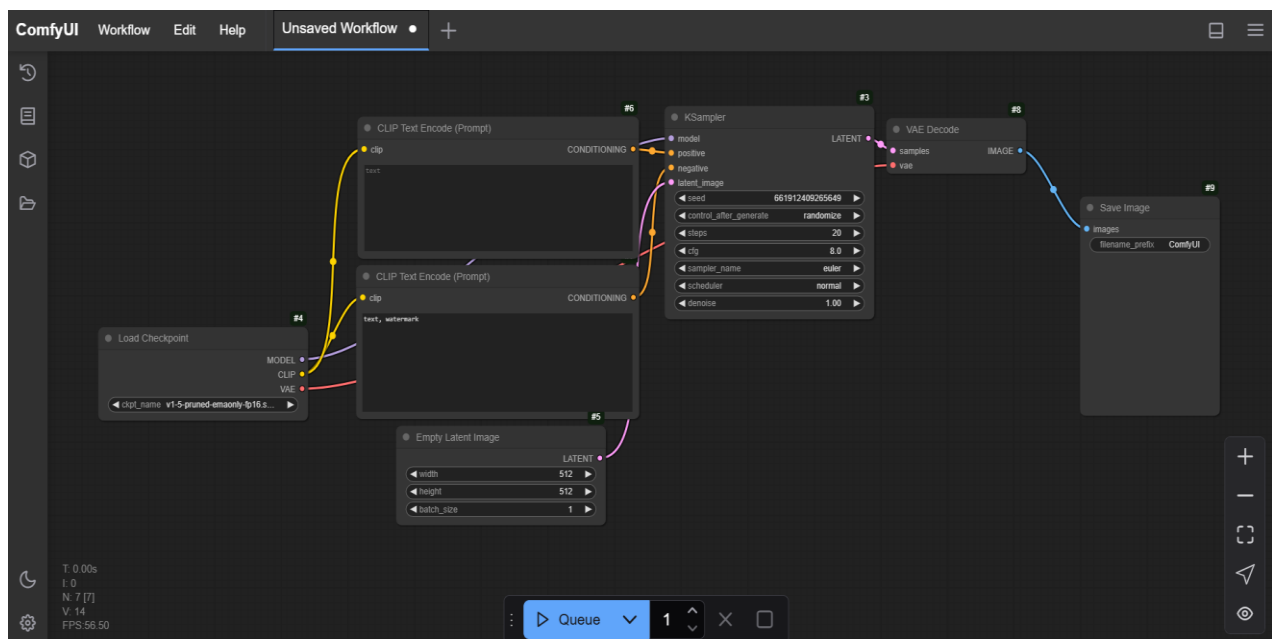


Figure 1: (Illustration of the sequential flow of image generation in Stable Diffusion using ComfyUI.)

1. Load Checkpoint

Purpose: Loads the Stable Diffusion model, which includes:

- The **UNet model** (for noise prediction)
- The **CLIP model** (for text encoding)
- The **VAE model** (for decoding the latent space into an image)

Inputs & Outputs

- **ckpt_name:** 526Mix_v145.safetensors (This is the Stable Diffusion model being used.)
- **Outputs:**
 - **MODEL** (purple): Goes to KSampler, representing the trained model used for image generation.
 - **CLIP** (yellow): Goes to CLIP Text Encode (Prompt), representing the text encoder for prompt processing.
 - **VAE** (red): Goes to VAE Decode, which converts the latent representation back into an image.

2. CLIP Text Encode (Prompt) – Positive Prompt

Purpose: Encodes the **positive text prompt** into an embedding that guides the image generation.

Inputs & Outputs

- **Prompt text:** "beautiful scenery nature glass bottle landscape, purple galaxy bottle"
- **clip (yellow):** Receives input from Load Checkpoint (CLIP model).
- **CONDITIONING (orange):** Connects to the **positive** input of KSampler, influencing the image towards this description.

Role: This guides the model to generate an image that aligns with the given description.

3. CLIP Text Encode (Prompt) – Negative Prompt

Purpose: Encodes the **negative text prompt** into an embedding to suppress unwanted elements.

Inputs & Outputs

- **Prompt text:** "text, watermark" (This ensures the model avoids adding text or watermarks.)
- **clip (yellow):** Receives input from Load Checkpoint (CLIP model).

- **CONDITIONING (orange)**: Connects to the **negative** input of KSampler, reducing the likelihood of undesired elements.

Role: Helps refine the output by preventing unwanted features.

4. Empty Latent Image

Purpose: Generates a blank latent tensor to act as the starting point for the image generation.

Inputs & Outputs

- **width:** 512
- **height:** 512
- **batch_size:** 1
- **LATENT (pink output)**: Connects to latent_image input of KSampler.

Role: This acts as an empty canvas where the model will apply noise and refine it through diffusion.

5. KSampler

Purpose: The core **diffusion process** where the model generates the image from noise.

Inputs

- **MODEL (purple)**: Receives input from Load Checkpoint, providing the Stable Diffusion model.
- **positive (orange)**: Receives input from CLIP Text Encode (Positive Prompt), guiding what the image should include.
- **negative (orange)**: Receives input from CLIP Text Encode (Negative Prompt), guiding what to exclude.
- **latent_image (pink)**: Receives input from Empty Latent Image, defining the starting latent space.

Key Parameters

- **Seed:** 156680208700286

- Controls randomness; changing it generates a different image.
- **Steps:** 20
 - More steps improve quality but increase processing time.
- **CFG (Classifier Free Guidance):** 8.0
 - Higher values enforce the prompt more, lower values allow more creativity.
- **Sampler Name:** Euler
 - Determines how the noise is refined; Euler is fast and works well for most cases.
- **Scheduler:** Normal
 - Defines the noise removal strategy.
- **Denoise:** 1.0
 - Controls how much denoising is applied; 1.0 applies full diffusion.

Outputs

- **LATENT (pink output):** Connects to VAE Decode, carrying the processed latent representation of the image.

Role: This is the primary node that converts text prompts into latent noise, refining it step by step to create an image.

6. VAE Decode

Purpose: Converts the **latent image** from `K

Sampler` into a visible RGB image.

Inputs & Outputs

- **samples (pink):** Receives input from KSampler, which contains the latent image representation.
- **vae (red):** Receives input from Load Checkpoint, providing the **Variational Autoencoder (VAE)** model used for decoding.

Role

- The **VAE model** converts the latent space data back into a real-world image.
- The model learns a **compressed representation of images**, making the process efficient.

Output

- **IMAGE (green output)**: Connects to Preview Image, representing the final image.

7. Preview Image

Purpose: Displays the final generated image.

Inputs & Outputs

- **image (green input)**: Receives input from VAE Decode, showing the final image on the ComfyUI interface.

Role: This is the last step, where the final image is rendered and ready for saving or further modification.

3.1.2 Step-by-Step Image Generation Process

1. **Load Model** → Loads Stable Diffusion model components (UNet, CLIP, VAE).
2. **Process Prompts**
 - Encode **positive prompt** (beautiful scenery...).
 - Encode **negative prompt** (text, watermark).
3. **Prepare Latent Space** → Generates an **empty latent tensor** (512x512).
4. **KSampler Applies Diffusion**
 - Uses **positive** and **negative** prompts.
 - Runs the **Euler sampler** for 20 steps.
 - Uses **CFG scale = 8.0** for prompt adherence.
5. **VAE Decodes Image** → Converts the latent representation into an RGB image.
6. **Preview Image** → Displays the final generated image.

3.1.3 Data Processing & Model Implementation

▪ Acquisition & Preprocessing

- Unlike conventional ML projects, **Stable Diffusion does not require direct training data.**
- Instead, **pre-trained weights** are used for generating images, with **ComfyUI facilitating workflow-based experimentation.**
- The model operates in **latent space**, **reducing computation costs** while maintaining high image fidelity.

▪ Understanding Forward & Reverse Diffusion

○ Forward Diffusion (Noise Addition Process):

1. An image is progressively **corrupted by Gaussian noise** over multiple timesteps (T).
2. The noise follows a Markov chain, ensuring gradual degradation of details.

○ Reverse Diffusion (Denoising Process):

1. U-Net takes a noisy latent vector as input and predicts the **noise residual** at each timestep.
2. The predicted noise is **subtracted from the input**, gradually refining the image back to its original form.
3. This process is repeated iteratively until a **clear and coherent image** emerges.

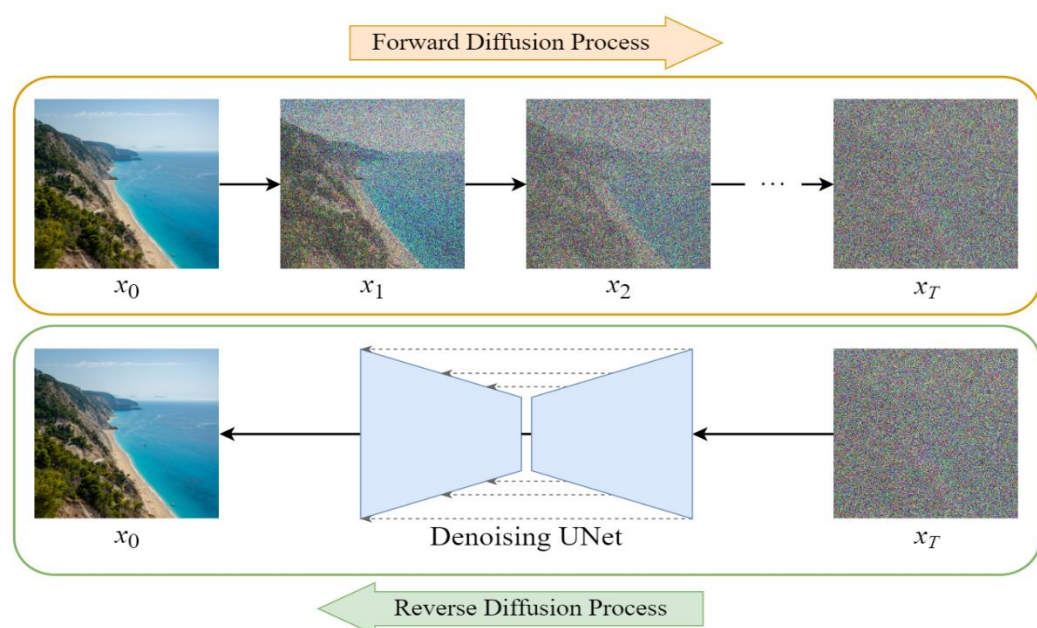


Figure 2: (Diagram illustrating forward and reverse diffusion steps in Stable Diffusion.)

3.2 Requirement Specification

3.2.1 Hardware Requirements:

- GPU: NVIDIA graphics card with at least 4GB VRAM, preferably an RTX 3060 or higher.
- CPU: Modern processor like Intel i5 or Ryzen 5
- RAM: 16GB or more
- Storage: Enough space to store AI models and generated images

3.2.2 Software Requirements:

- Operating System: Windows 10/11 or Linux
- ComfyUI application: Downloadable from the official ComfyUI repository
- CUDA drivers: Necessary for utilizing the NVIDIA GPU
- Stable Diffusion model version: v1-5-pruned-emaonly-fp16.safetensors from Hugging Face model hub

CHAPTER 4

Implementation and Result

4.1 Snap Shots of Result:

To demonstrate how the ComfyUI model works, different types of prompts are given to show how Stable Diffusion reacts to various styles, themes, and constraints. Below are different categories of prompts along with what they showcase:

4.1.1 Simple Object-Based Prompts (Baseline)

Showcase: How well the model generates basic objects

Prompt: "A red apple on a wooden table"

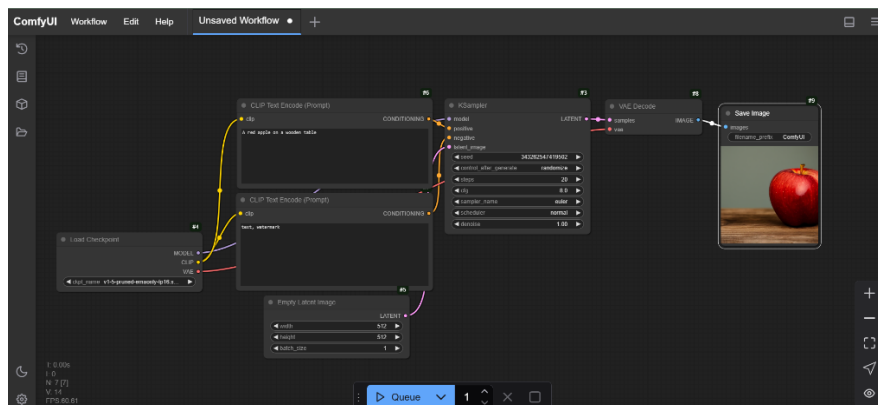


Figure 2: (Illustrating the red apple prompt)

The model generates a **realistic apple** with appropriate textures and shadows.

Accuracy: High for well-defined, simple objects.

4.1.2 Style-Based Prompts

Showcase: How different artistic styles affect the image

Prompt: "A cozy cottage in the forest, Studio Ghibli style"



Figure 2: (Illustrating the cozy cottage prompt)

The model applies **soft lighting, pastel colors, and an animated feel** to match the Ghibli aesthetic.

Accuracy: Good at mimicking well-known artistic styles.

4.1.3 Environmental Change Prompts

Showcase: How well the model generates detailed multi-object compositions

Prompt: "A futuristic city covered in heavy fog and neon lights"



Figure 2: (Illustrating futuristic city prompt)

The model **adds cyberpunk elements like neon colors, dark tones, and futuristic details**.

Accuracy: Decent for complex characters, though fine details like tattoos may not always be sharp.

4.1.4 Character-Based Prompts

Showcase: How well the model generates human-like characters

Prompt: "A cyberpunk warrior with neon tattoos and a robotic arm"



Figure 2: (Illustrating cyberpunk warrior prompt)

The model **adds cyberpunk elements like neon colors, dark tones, and futuristic details.**

Accuracy: Decent for complex characters, though fine details like tattoos may not always be sharp.

4.1.5 Scene Complexity & Composition

Showcase: How well the model generates detailed multi-object compositions

Prompt: "A busy marketplace with colorful stalls, people interacting"



Figure 2: (Illustrating busy marketplace prompt)

The model generates the image that looks something like a market place.

Accuracy: The **people have unnatural poses** or objects may appear in odd positions.

4.1.6 Multi-Prompt Blending

Showcase: How well the model mixes different concepts

Prompt: "A dragon made of fire and ice, glowing eyes"



Figure 2: (Illustrating dragon prompt)

When multiple unrelated elements are included in a single prompt, the model may **blend** them incorrectly.

Accuracy: The fire and ice might mix strangely instead of being separate features.

4.1.7 Multiple Structured Constraints

Showcase: How well the model handles multiple structured constraints at once.

Prompt: "A giant, ultra-detailed, perfectly spherical glass orb floating in the sky, reflecting a sunset over a vast futuristic city. The city has towering triangular skyscrapers with symmetrical neon-lit patterns. The perspective is wide-angle, capturing depth with a cinematic 16:9 aspect ratio. The scene is illuminated by golden-hour lighting, with soft reflections on the orb's surface. The ground features a hexagonal grid structure. The background fades into a misty horizon, adding a sense of depth and dimension."



Figure 2: (Illustrating spherical glass orb prompt)

The model **captures reflections, depth, and scale.**

Accuracy: High in creating large-scale environments, but the orb may have slight deformations rather than a perfect sphere.

4.2 GitHub Link:

https://github.com/KusumaPriya15/ComfyUI_Model

CHAPTER 5

Discussion and Conclusion

5.1 Future Work:

While this project successfully demonstrates **image generation using Stable Diffusion and ComfyUI**, there are areas that can be improved:

- **Faster Image Generation**
 - Right now, the model **takes time to generate images** because it follows multiple denoising steps.
 - **How to Improve:** Using **optimized diffusion techniques** to speed up image generation without losing quality.
- **Better Match Between Text and Images**
 - Sometimes, the images do not fully match the **exact meaning of the prompt**.
 - **How to Improve:**
 - Fine-tune **CLIP embeddings** to make images more accurate.
 - Train the model on **larger datasets** with better text descriptions.
- **More Control Over Image Style**
 - Users may want more control over **art style, lighting, and composition**.
 - **How to Improve:**
 - Use **additional control settings** in ComfyUI.
 - Experiment with **custom AI models for different artistic styles**.
- **Extending to Video Generation**
 - Right now, Stable Diffusion works **only for images**.
 - **How to Improve:**

- Explore **Stable Video Diffusion** to create short AI-generated videos.
- Use **frame interpolation** to smooth video motion.

Future Goal: Make AI-generated images **faster, more accurate, and customizable**, while exploring **video generation**.

5.2 Conclusion:

This project explored how Stable Diffusion and ComfyUI can generate images using AI. The main achievements include:

- ✓ Understanding how Stable Diffusion works (adding/removing noise to create images).
- ✓ Learning how U-Net helps refine images step by step.
- ✓ Testing different prompts, settings, and diffusion steps to improve results.
- ✓ Using ComfyUI for an easy, customizable AI workflow.
- ✓ Evaluating image quality using CLIP scores and FID metrics.

Key Takeaways:

- Stable Diffusion is a powerful AI tool for generating realistic images.
- ComfyUI makes it easier to experiment with settings and improve results.
- Future improvements can make AI image generation faster and more detailed.

This project helps in understanding how AI creates images and sets the stage for future AI-powered creativity.

REFERENCES

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 - Introduces **Latent Diffusion Models (LDMs)**, which improve image synthesis by performing the diffusion process in a lower-dimensional latent space, making AI-generated images more efficient and high-resolution.
- [2]. Ho, J., Jain, A., & Abbeel, P. (2020). "**Denoising Diffusion Probabilistic Models.**" Advances in Neural Information Processing Systems (NeurIPS).
 - Proposes **Denoising Diffusion Probabilistic Models (DDPMs)**, a generative modeling approach that gradually removes noise from random inputs to generate realistic images, laying the foundation for modern diffusion models like Stable Diffusion.